

STAT 412 QZ07 — Question 6

Problem

A research report claims that mice with an average life span of 32 months will live to be about 41 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that the true mean $\mu < 41$ if 70 mice on this diet have an average life of 40 months with a standard deviation of 3.4 months? Use a P-value in the conclusion.

Solution

The question asks whether the mean is less than 41:

$$H_0 : \mu = 41, \quad H_1 : \mu < 41.$$

This is choice F.

Use a one-sample t test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{40 - 41}{3.4/\sqrt{70}} = -2.46.$$

The degrees of freedom are $df = 70 - 1 = 69$. For a left-tailed test,

$$P = P(T_{69} \leq -2.46) \approx 0.0082.$$

From the t table, this falls in the range

$$0.0075 < P\text{-value} < 0.01.$$

Since the P-value is small, reject H_0 .

Final Answer

Choice F. $t = -2.46$; $0.0075 < P\text{-value} < 0.01$. Reject H_0 ; there is sufficient evidence that the mice will live to be less than 41 months old.